

POIS Question Bank

February 2024

Problem 1. Consider a modification of the substitution cipher, where instead of applying only the substitution, we first apply a substitution and then apply a shift cipher on the substituted values. Give a formal description of this scheme and show how to break the substitute and shift cipher.

Problem 2. Consider the double Vigenère cipher, where we choose two keys k_1, k_2 with different periods $t_1, t_2 \in [t_{max}]$, and then encrypt a message in the following way: first encrypt the message using the Vigenère cipher with k_1 to obtain an intermediate ciphertext c' , and then encrypt c' using the Vigenère cipher with k_2 to obtain the final ciphertext c . Show how to break the double Vigenère cipher.

Problem 3. Assume the adversary has the ability to obtain ciphertexts for arbitrary plaintexts, i.e., it can choose a message and receive an encryption of the message without knowing the secret key. Attacks that leverage this information are known as *chosen plaintext attacks*.

- (a) Using chosen plaintext attacks, show how to learn the secret key for the shift, substitution, and Vigenère ciphers. Your attacks should each use only a single plaintext. What is the smallest plaintext length that suffices for each attack?
- (b) For the Vigenère cipher, please consider two cases: (a) when the period t is known; (b) when the period t is unknown but an upper bound t_{max} on t is known. For the latter case, an asymptotic analysis of the required plaintext length suffices, i.e., an upper bound on the required plaintext length.

Problem 4. Negligible or not?

- (a) Let $f, g : \mathbb{N} \rightarrow \mathbb{R}$ be negligible functions, let $p : \mathbb{N} \rightarrow \mathbb{R}$ be a polynomial such that $p(n) > 0$ for all $n \in \mathbb{N}$.
 - 1. Define $h : \mathbb{N} \rightarrow \mathbb{R}$ as $h(n) = f(n) + g(n)$.
 - 2. Define $h : \mathbb{N} \rightarrow \mathbb{R}$ as $h(n) = f(n) \cdot p(n)$.
 - 3. $f(n) := f'(n) \cdot p(n)$, for some negligible function $f'(n)$ and some polynomial function $p(n)$. Is such a $f(n)$ always negligible?
- (b) For each of the following functions below, either prove or disprove that it is negligible.
 - 1. $f(n) := 1/2^{100 \log n}$.
 - 2. $f(n) := 1/(\log n)^{\log n}$.
 - 3. $f(n) = n^{-100} + 2^{-n}$
 - 4. $f(n) = 1.01^{-n}$
 - 5. $f(n) = 2^{-(\log n)^2}$
 - 6. $f(n) = 2^{-\sqrt{n}}$
 - 7. $f(n) = 2^{-\sqrt{\log n}}$
 - 8. $f(n) = e^{-\log^3 n} + e^{-\log^2 n} + e^{-\log n}$

Problem 5. PRG or not? Let $G : \{0, 1\}^{2n} \rightarrow \{0, 1\}^{2n+1}$ be a pseudorandom generator (PRG). For each part below, either prove or disprove that $G' : \{0, 1\}^{2n} \rightarrow \{0, 1\}^{2n+1}$ is *necessarily* a PRG no matter which PRG G is used.

- (a) $G'(x) := G(\pi(x))$ for, where $\pi : \{0, 1\}^{2n} \rightarrow \{0, 1\}^{2n}$ is any $\text{poly}(n)$ -time computable bijective function. (You may not assume that π^{-1} is $\text{poly}(n)$ -time computable.)
- (b) $G'(x||y) := G(x||x \oplus y)$, where $|x| = |y| = n$. (**Note:** Here, and throughout the course, $x||y$ refers to the concatenation of two strings x and y .)
- (c) $G'(x||y) := G(x||0^n) \oplus G(0^n||y)$, where $|x| = |y| = n$. (**Note:** Here, and throughout the course, 0^n and 1^n denote the string of 0s and 1s, respectively, of length n .)
- (d) $G'(x||y) := G(x||y) \oplus (x||0^{n+1})$, where $|x| = |y| = n$.

Problem 6. An encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ over message space $\mathcal{M}$ and ciphertext space $\mathcal{C}$ is said to be 2-time perfectly secure if for any $(m_1, m_2) \in \mathcal{M} \times \mathcal{M}$ and $(m'_1, m'_2) \in \mathcal{M} \times \mathcal{M}$ such that $m_1 \neq m_2$ and $m'_1 \neq m'_2$ and for any $c_1, c_2 \in \mathcal{C}$ the following holds:

$$\Pr[\text{Enc}(K, m_1) = c_1 \wedge \text{Enc}(K, m_2) = c_2] = \Pr[\text{Enc}(K, m'_1) = c_1 \wedge \text{Enc}(K, m'_2) = c_2].$$

Note that in the above definition the key K is the same for encrypting m_1, m_2 (resp. m'_1, m'_2). Consider the following encryption scheme for the message space $\mathbb{Z}_{23}$.

- Gen:** Sample two elements $a \xleftarrow{\$} \mathbb{Z}_{23}$ and $b \xleftarrow{\$} \mathbb{Z}_{23}$.
- Enc** $((a, b), m)$: Output $c = a \cdot m + b \pmod{23}$.
- Dec** $((a, b), c)$: Compute $m = (c - b) \cdot a^{-1} \pmod{23}$ if a is invertible. Otherwise, output error.

Show the following.

- (a) Prove that for any message $m \in \mathbb{Z}_{23}$,

$$\Pr[\text{Dec}(K, \text{Enc}(K, m)) = m] = \frac{22}{23}$$

- (b) Prove that this is 2-time secure.

Problem 7. Let $f : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a pseudorandom function (PRF). For the functions f' below, either prove that f' is a PRF (for all choices of f), or prove that f' is not a PRF.

- (a) $f'_k(x) := f_k(0||x)||f_k(1||x)$.
- (b) $f'_k(x) := f_k(0||x)||f_k(x||1)$.

Problem 8. Let F be a PRF defined over (K, X, Y) , where $X = \{0, \dots, N-1\}$ and $Y = \{0, 1\}^n$, where N is super-poly. For poly-bounded $\ell \geq 1$, consider the PRF F' defined over (K, X, Y^ℓ) as follows:

$$F'(k, x) := (F(k, x), F(k, x+1 \pmod{N}), \dots, F(k, x+\ell-1 \pmod{N})).$$

- (a) Show that F' is a weakly secure PRF,
- (b) Proof that randomized counter mode is CPA secure.

Problem 9. Assume that one-way functions exist. $b : \{0, 1\}^* \rightarrow \{0, 1\}$ is a hard-core predicate of a one-way function $f(\cdot)$ if $b(x)$ is efficiently computable given x and there exists a negligible function v such that for every PPT adversary A and for every n :

$$\Pr[x \leftarrow \{0, 1\}^n : A(f(x)) = b(x)] \leq \frac{1}{2} + v(n).$$

- (a) Construct an one-way function $f : \{0, 1\}^n \rightarrow \{0, 1\}^m$ for input $x = (x_1, x_2, \dots, x_n)$ such that the first bit x_1 is not hardcore.
- (b) A polynomial time-computable predicate $b : \{0, 1\}^n \rightarrow \{0, 1\}$ is called a *universal* hard-core predicate if for *every* one-way function $f : \{0, 1\}^n \rightarrow \{0, 1\}^m$, b is hardcore. Prove that there is no universal hardcore predicate.
- (c) Could there be a one-way function f for which none of the individual bits of the input are hardcore? It turns out that there are such beasts. Construct a one-way function $f : \{0, 1\}^n \rightarrow \{0, 1\}^m$ for which none of the predicates $b_i(x_1, \dots, x_n) = x_i$ are hardcore.

Problem 10. Show that the basic CBC-MAC construction is not secure when considering messages of different lengths.

•**Basic CBC-MAC Construction:**

- Gen(1^n):** upon input 1^n , choose a uniformly distributed string $k \leftarrow \{0, 1\}^n$.
- Mac $_k(m)$:** upon input key $k \in \{0, 1\}^n$ and a message m of length $\ell \cdot n$, do the following:
 1. Denote $m = m_1, \dots, m_\ell$ where each m_i is of length n , and set $t_0 = 0^n$.
 2. For $i = 1$ to ℓ , set $t_i \leftarrow F_k(t_{i-1} \oplus m_i)$ where $F : \{0, 1\}^* \times \{0, 1\}^* \rightarrow \{0, 1\}^*$ is a function.
 3. Output t_ℓ .

Verification Algorithm:

- Vrfy $_k(m, t)$:** upon input key $k \in \{0, 1\}^n$, a message m of length $\ell \cdot n$, and a tag t of length n , output 1 if and only if $t = \text{Mac}_k(m)$.

Problem 11. Suppose $(\text{Gen}, \text{Enc}, \text{Dec})$ is a CPA-secure encryption scheme that encrypts messages belonging to a field $\mathcal{F}$. Construct a new encryption scheme as follows:

- Gen $_1(1^n)$:** samples $k' \xleftarrow{\$} \text{Gen}(1^n)$. Then it samples p , a random degree- d polynomial over the field $\mathcal{F}$. The key k for this encryption scheme is the tuple of these values:

$$k = (k', p)$$

(here p refers to the description of the polynomial).

- Enc $_1(k, m)$** computes and outputs $c = \text{Enc}(k', m) \parallel p(m)$.
- Dec $_1(k, c)$** just runs $\text{Dec}(k', \cdot)$ on the first part of the ciphertext.

In the CPA security experiment, what is the minimum number of queries to the Enc_1 oracle that are needed to break the CPA security of the scheme $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$. (i.e) What is the minimum number of queries needed to figure out b given $\text{Enc}_1(k, mb)$?

Problem 12. Give complete details of how to construct Y from X where

- (a) X = One-way permutation, Y = pseudorandom generator
- (b) X = pseudorandom generator, Y = one-way function
- (c) X = pseudorandom generator, Y = pseudorandom function

Problem 13. Assuming pseudorandom functions exist, create a CPA-secure private key encryption scheme.

Problem 14. Prove that weak one-way functions exist if and only if strong one-way functions exist.

Problem 15. Let $\mathcal{F} : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a pseudorandom function (PRF) and let

$$\mathcal{G}(k, (x, y)) = \mathcal{F}(k, x) \oplus \mathcal{F}(k, y)$$

Prove that $\mathcal{G}$ is not a pseudorandom function.

Problem 16. Let $\mathcal{F} : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a pseudorandom function (PRF). Prove that the following function $\mathcal{H}$ is also a pseudorandom function:

$$\mathcal{H}(k, x) = \mathcal{F}(k, x) \oplus x$$

Problem 17. Let $(\text{Gen}, \text{Enc}, \text{Dec})$ be a CPA-secure encryption scheme. Below, we will construct another encryption scheme and prove that it is also CPA-secure. In the encryption scheme below, let the message m belong to $\{0, 1\}^n$.

- $\text{Gen}_1(1^n)$: Sample the key as follows: $k \leftarrow \text{Gen}(1^n)$
- $\text{Enc}_1(k, m)$: Sample $r \leftarrow \{0, 1\}^n$ uniformly at random. Then compute $c_0 := \text{Enc}(k, r)$ and $c_1 := r \oplus m$. Output the ciphertext $c = (c_0, c_1)$.
- $\text{Dec}_1(k, (c_0, c_1))$: Unspecified.

- (a) Fill in the decryption algorithm $\text{Dec}_1(k, (c_0, c_1))$ so that every ciphertext is decrypted correctly.
- (b) Prove that $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$ satisfies CPA security.

Problem 18. In this problem we propose a definition of perfect secrecy for the encryption of two messages. We will prove that this definition cannot be satisfied by any encryption scheme.

Notation: We consider distributions over pairs of messages from the message space $\mathcal{M}$; we let M_1 and M_2 be random variables denoting the first and second message, respectively. (We stress that these random variables are not assumed to be independent.) Encryption works as follows:

- Generate a (single) key k , sample a pair of messages (m_1, m_2) according to the given distribution.
- Compute ciphertexts $c_1 \leftarrow \text{Enc}(k, m_1)$ and $c_2 \leftarrow \text{Enc}(k, m_2)$. This induces a distribution over pairs of ciphertexts, and we let C_1 and C_2 be the corresponding random variables.

Proposed Definition: Let us say that an encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ is *perfectly secret for two messages* if for all distributions over $\mathcal{M} \times \mathcal{M}$, all $(m_1, m_2) \in \mathcal{M} \times \mathcal{M}$ and all ciphertexts $(c_1, c_2) \in \mathcal{C} \times \mathcal{C}$ for which $\Pr[C_1 = c_1 \wedge C_2 = c_2] > 0$,

$$\Pr[M_1 = m_1 \wedge M_2 = m_2 | C_1 = c_1 \wedge C_2 = c_2] = \Pr[M_1 = m_1 \wedge M_2 = m_2]$$

Question: Prove that no encryption scheme can satisfy the definition above. (You may assume that the encryption scheme satisfies perfect correctness).

Problem 19. Show that the CBC and Counter modes of encryption are not CCA-secure.

Problem 20. Consider a weaker definition of CPA security where in the indistinguishability experiment the adversary $\mathcal{A}$ is not given oracle access to $\text{Enc}_k(\cdot)$ after choosing m_0, m_1 . That is, $\mathcal{A}$ can only query $\text{Enc}_k(\cdot)$ in phase 1, but not in phase 2. We call this definition weak-CPA-security. Prove that weak-CPA-security is equivalent to CPA-security (i.e., Definition 3.22 in the textbook).

Problem 21. Define functions $f, g : \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$, and let $g(n) = 2^{-f(n)}$.

- (a) Prove that if $f(n) = \omega(\log n)$, then $g(n)$ is negligible.
- (b) Prove that if $f(n) = O(\log n)$, then $g(n)$ is non-negligible.

Problem 22. Let f be a length-preserving one-way function. For each of the following parts, either prove that f_i is necessarily a one-way function or show a counterexample (namely, a length-preserving one-way function f for which f_i is *not* one-way.) Your counterexamples must rely only on the *existence* of one-way functions.

- (a) $f_0(x) = f(f(x))$
- (b) $f_1(x, y) := f(x) \| f(x \oplus y)$, where the length of x and y is the same, i.e., $|x| = |y|$.
- (c) $f_2(x) := (f(x) \| x_{[1:\log |x|]})$, where the notation $y_{[1:\ell]}$ denotes the string y restricted to its first ℓ bits.
- (d) $f_3(x) := f(x)_{[1:|x|-1]}$.
- (e) $f_4(x) := f(x) \oplus x$.

Problem 23. Let $\{\mathcal{F}_n\}_{n \in \mathbb{N}}$ where

$$\mathcal{F}_n = \{f_k : \{0, 1\}^\ell \rightarrow \{0, 1\}^\ell\}_{k \in \{0, 1\}^\ell}$$

is a pseudo-random function (PRF) family. For each of the following parts, either prove that the function family $\{\mathcal{F}'_n\}_{n \in \mathbb{N}}$ where

$$\mathcal{F}'_n = \{f'_k : \{0, 1\}^{2\ell} \rightarrow \{0, 1\}^\ell\}_{k \in \{0, 1\}^\ell}$$

is a pseudorandom function family or provide a polynomial-time attack showing that it is not.

- (a) $f'_k(x, y) = f_k(x) \oplus f_k(y)$.
- (b) $f'_k(x, y) = f_k(x \oplus y)$.
- (c) $f'_k(x, y) = f_{k \oplus x}(y)$.
- (d) $f'_k(x, y) = f_{f_k(x)}(y)$.

Problem 24. Let (Enc, Dec) be a computationally indistinguishable symmetric encryption scheme and MAC an existentially unforgeable message authentication scheme under Chosen Message Attack (CMA): i.e., any adversary who can request MACs on messages $M_1, M_2, \dots$ of his choice cannot generate a valid forgery on a new message M' .

Suppose Alice and Bob share two independent keys K_1, K_2 (generated at random using the respective generation algorithms) for privacy and authentication, respectively. They want to exchange messages M in a private and authenticated way. Consider sending each of the following as a means to this end, and argue whether it is secure or not, where secure means both computationally indistinguishable and existentially unforgeable under CMA.

- (a) $M, \text{MAC}_{K_2}(\text{Enc}_{K_1}(M))$
- (b) $\text{Enc}_{K_1}(M, \text{MAC}_{K_2}(M))$
- (c) $\text{Enc}_{K_1}(M), \text{MAC}_{K_2}(M)$
- (d) $\text{Enc}_{K_1}(M), \text{MAC}_{K_2}(\text{Enc}_{K_1}(M))$
- (e) $\text{Enc}_{K_1}(M, A)$, where A denotes the identity of Alice. Bob decrypts the ciphertext and checks that the second half of the plaintext is A .